

Recall from a previous class:

- A set of vectors  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

- Let  $A = [\vec{a}_1 \dots \vec{a}_n]$ , then  $\{\vec{a}_1, \dots, \vec{a}_n\}$  are linearly independent if and only if  $A\vec{x} = \vec{0}$  has only the trivial solution.
- **Theorem 7:** A set  $S = \{\vec{v}_1, \dots, \vec{v}_p\}$  of two or more vectors is linearly dependent if and only if one of the vectors of  $S$  is a linearly combination of the others.

Moreover if  $\vec{v}_1 \neq \vec{0}$ , then some  $\vec{v}_j$  for  $j > 1$  is a linear combination of  $\vec{v}_1, \dots, \vec{v}_{j-1}$ .

- **Theorem 8:** Any set  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is linearly dependent if  $p > n$ .

Define a transformation  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  to be a function from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ . Recall we call  $\mathbb{R}^n$  the domain and  $\mathbb{R}^m$  the codomain. The set of outputs,  $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$ , is the image or the range of  $T$ .

One common example of a transformation  $T$  is given by  $T(\vec{x}) = A\vec{x}$  where  $A$  is an  $m$  by  $n$  matrix

**Definition 1.** A transformation  $T$  is **linear** if

1.  $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$  for all  $\vec{u}, \vec{v}$  in  $\mathbb{R}^n$ ,
2.  $T(c\vec{u}) = cT(\vec{u})$  for all  $\vec{u}$  in  $\mathbb{R}^n$  and  $c$  in  $\mathbb{R}$ .

### 1. Properties of Linear Transformations.

(a) Prove if  $T$  is linear, then  $T(\vec{0}) = \vec{0}$ .

(b)  $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$ .

2. Let  $T(\vec{x}) = A\vec{x}$  where  $A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \\ 2 & -2 \end{bmatrix}$  and let  $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$ .

(a) Is  $\vec{b}$  in the range of  $T$ ?

(b) What is the range of  $T$ ?

(c) Is  $T$  surjective?

(d) Is the transformation  $T$  injective?

3. Let  $T(\vec{x}) = A\vec{x}$  where  $A = \begin{bmatrix} 1 & 0 & -3 \\ -3 & 1 & 6 \\ 2 & -2 & -1 \end{bmatrix}$  and let  $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$ . Is  $\vec{b}$  in the range of  $T$ ?

4. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear and  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  be a linearly dependent set. Is  $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$  linearly dependent?

5. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear and  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  be a linearly **in**dependent set. Is  $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$  linearly **in**dependent?